Hotstar (Disney+ Hotstar / JioHotstar): Cloud Scaling for Record Cricket Streaming

IT457 Cloud Computing Case Study

Week 1 Summary ·

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21 Oct 2026

1. Sources found

Primary (Hotstar engineering and AWS)

- **P1** "Scaling Is Not An Accident" (Hotstar blog, Apr 2018): pre-scaling with concurrency ladders instead of reactive autoscaling.
- **P2** "T For Tsunami: Dealing with traffic spikes" (Hotstar blog): spike handling and bitrate degradation.
- **P3** "Scaling the Hotstar Platform for 50M" (Oct 2019): EC2 to Kubernetes, custom autoscaler, "jettison" plan, compute savings.
- **P4** "Hotstar's Journey from EC2 to Containers": why static EC2/ASG scaling failed.
- **P5** "Scaling Infrastructure for Millions, Part 1": 2023 World Cup prep (NAT, node, IP and Kubernetes limits).
- **P6** "Datacenter Abstraction, Part 2" (Jun 2025): six EKS clusters, Envoy gateway, 61M+ outcome.
- **P7** G. Kamboj, "Scaling Hotstar.com for 25 million concurrent viewers" (CMY302, AWS re:Invent 2019): Hulk load testing, GameDays, chaos engineering. Read via the Classmethod session report; original video to be cited directly.
- **P8** "Saving Millions: Leveraging EC2 Spots at Scale", and **P9** "The Kafka Optimization That Saved Us $350K a Year" (Hotstar blog): cost levers.

Secondary (cross-checking only): ByteByteGo, System Design Newsletter, and press coverage of JioStar releases (Outlook Business, exchange4media). **Excluded as unverified:** GeeksforGeeks/TechAhead infrastructure-spec claims and personal blogs with no primary source.

2. Case study salient points

Problem. Live cricket traffic is event-driven and extremely spiky: notifications and big moments create a "tsunami" that arrives faster than instances can launch, and client retries make overload worse. Before the 2023 World Cup, peak was near 25M on two self-managed Kubernetes clusters, while modelling pointed to about 50M (P5, P6).

Strategy evolution

- **2018, pre-scaling:** reactive autoscaling was rejected as too slow; capacity was scaled ahead of events using static "ladders" based on estimated concurrency (P1, P3).

- **2019, Kubernetes:** a custom autoscaler, run in shadow mode before being trusted, delivered about 2x the concurrency with about 10x less compute (company claim, P3).
- **2023, managed EKS:** six EKS clusters grouped as one “data center”; an Envoy gateway replaced 200+ per-service load balancers that needed pre-warming (P6).

Defence layers. CDN as origin shock absorber (cacheable vs non-cacheable APIs, lean CDN domain, lower refresh rates); smart clients with jittered back-off; “jettisoning” non-essential features so video keeps playing; Hulk load tests, GameDays and chaos testing (P3, P5, P7).

Bottlenecks fixed (2023). NAT gateways moved from per-AZ to per-subnet; node scale-up in steps of 100–300 to avoid API server throttling; IP exhaustion solved with /19 subnets; services capped below 1,000 pods (P5).

Cost evidence. No total cloud bill is public. Quantified items: 10x compute reduction in 2019 (P3); spot instances for data and load-generation workloads (P7, P8); telemetry payloads cut by about 85%, saving close to $350K a year (P9).

Outcomes (peak concurrent viewers, company-reported)

Event	Peak
IPL 2018 final	10.3M
IPL 2019 final	18.6M
World Cup semi-final, Jul 2019	25.3M
World Cup 2023 final	59M
IPL 2025 final	55M
Champions Trophy 2025 final	61.2M

Only concurrency is quoted; “views” is a different metric. The 61.2M figure is JioStar’s own (6.12 crore); some outlets round it to 62M.

3. Gaps and next steps

Cost data is limited to percentage claims and one dollar figure, so this will be stated as a limitation. P7 needs cross-checking against the actual AWS video. **Week 3 exercise:** compare reactive autoscaling with pre-scaled capacity on an ALB and Auto Scaling Group under the same spike profile (ramp, spike, plateau, drop) using k6 or Locust, reporting p50/p95 latency, error rate, scale-out lag and cost per run.